

NeuralChat — End-to-End System Design · Section 3 of 3

Runtime & Operations — Flows, Security, Reliability & the Build

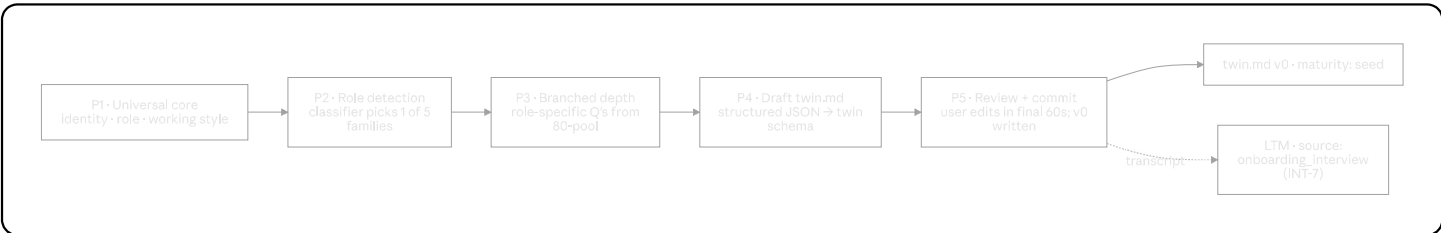
Doc ID	NE-TSD-NC-V2 · S3
Owner	Mansi Gambhir (VP AI Research)
Runtime	Hermes host · Pi-agents execute
Reads with	S1 (Foundation) · S2 (Intelligence)

Section 3 scope. The system *running*: the eight end-to-end flows that cross components, the four-layer tenant-isolation ACL, the full security posture, reliability/DR, compliance, and the dependency-ordered 90-day build with acceptance criteria and risks.

3.1 The eight end-to-end flows

Flow 1 — Onboarding seeds twin v0 (TRD §4.8)

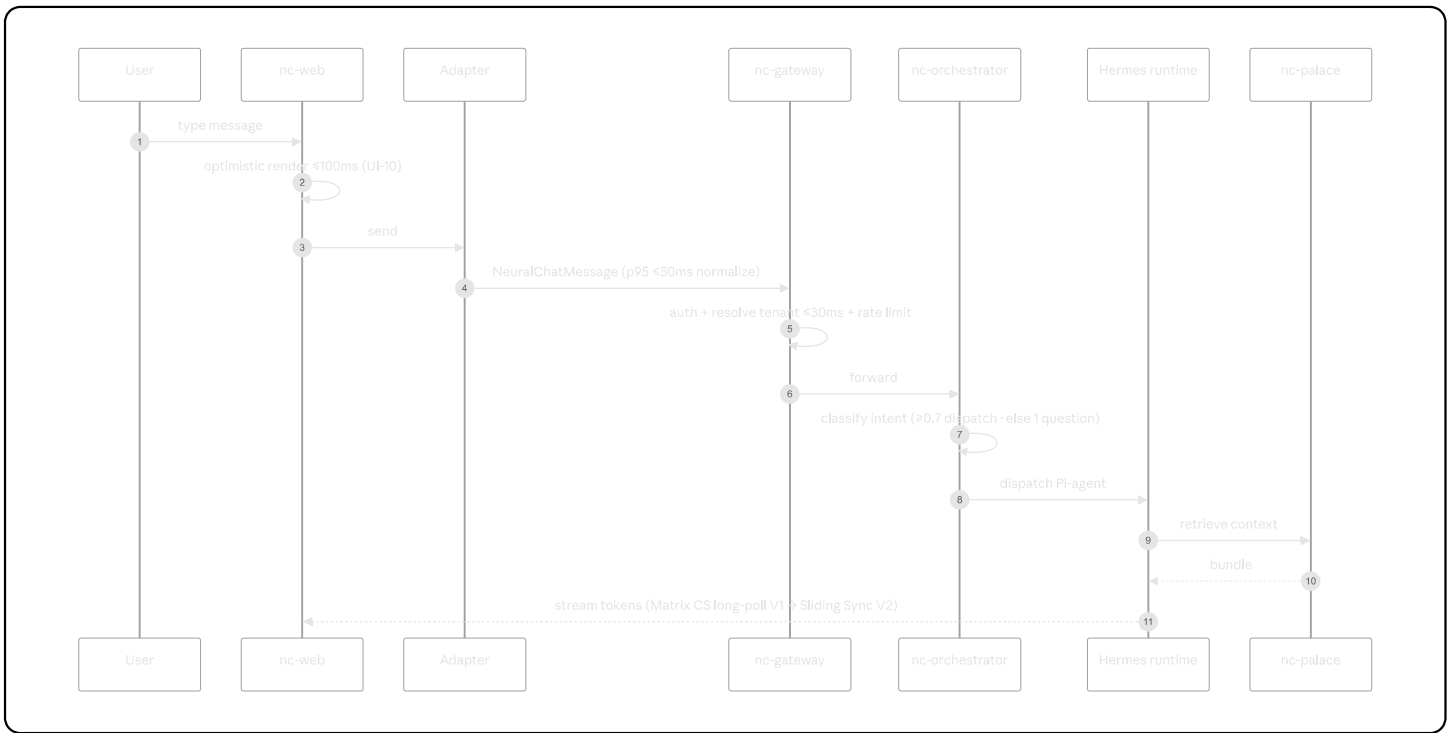
The **Interviewer Pi-agent** turns a ≤15-minute conversation into a structured twin file. Onboarding is the day-one self-model build.



- **INT-1/4** — 15-min wall-clock cap; each phase has min/max question count + time budget.
- **INT-8** — re-runnable on role change; replaces twin v0 but **preserves accumulated signals** via the Twin Curator.

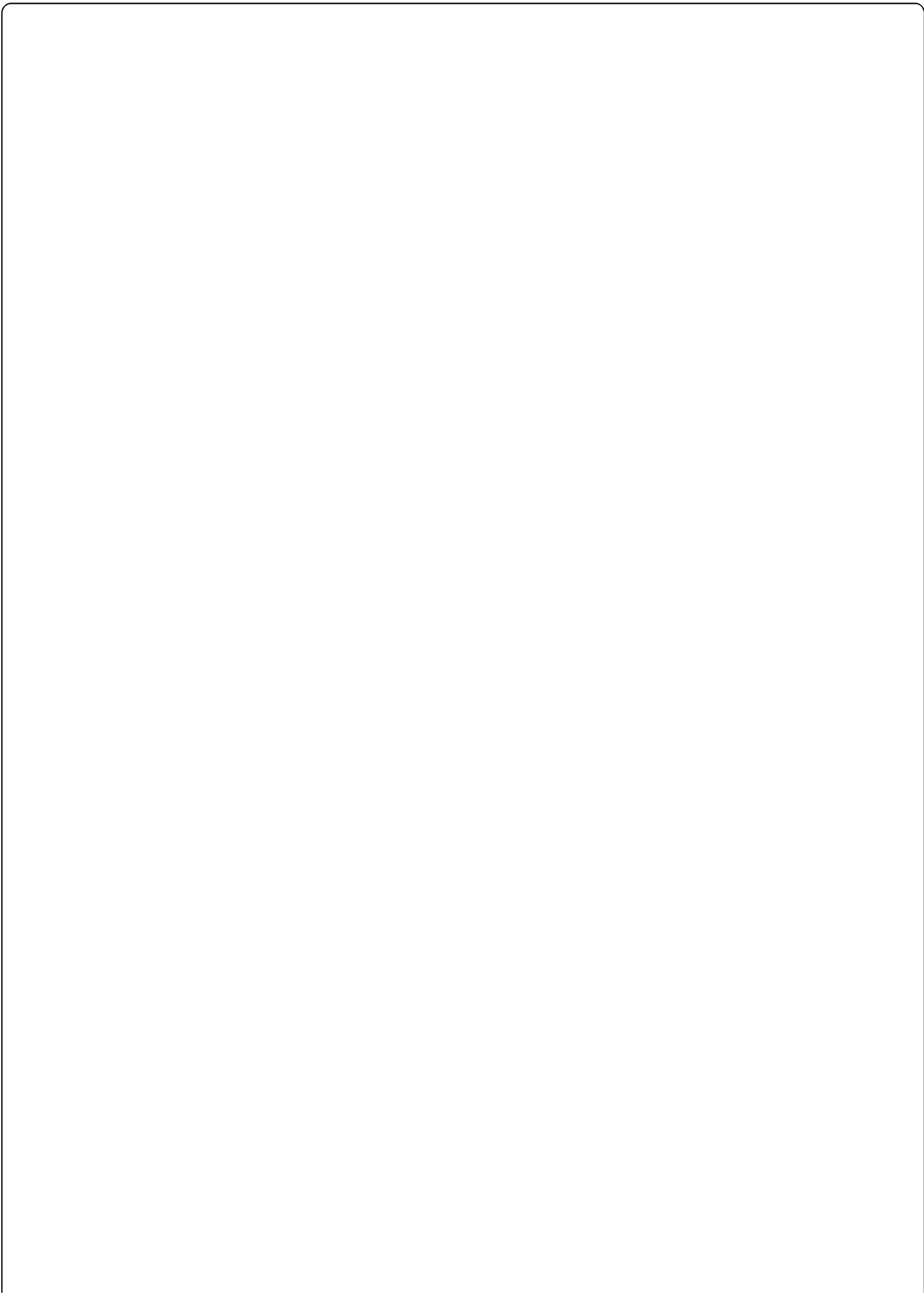
Flow 2 — A chat message round-trip (TRD §3.1, NFR-1)

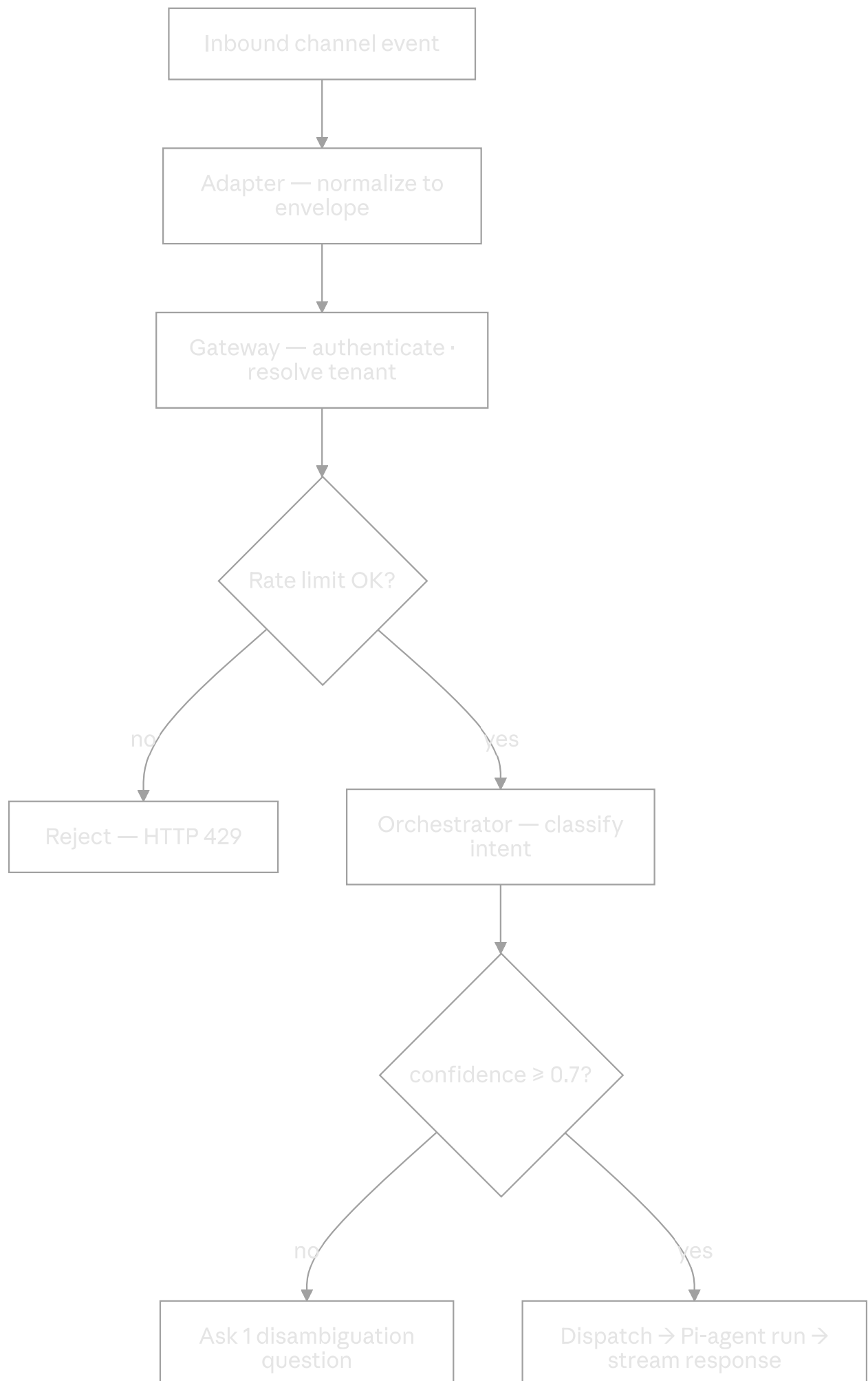
The latency-critical path. **Budget: p50 ≤ 800 ms, p95 ≤ 2 s** (excluding Pi-agent work).



Flow 3 — Request lifecycle, gate by gate

Two decision gates stand between an inbound message and a Pi-agent run.

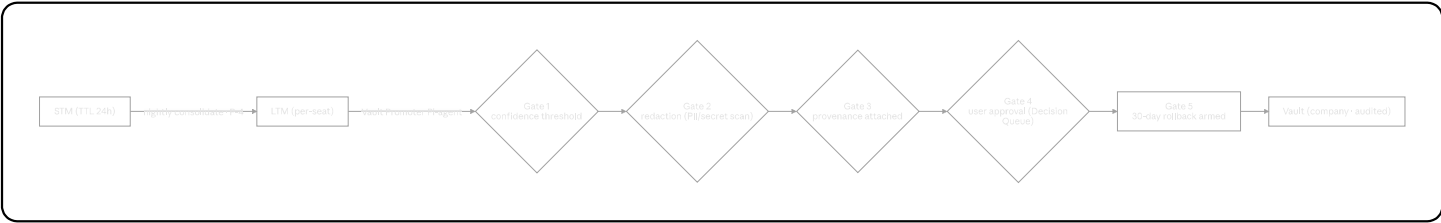




The internal **plan-execute-verify** sequence and **run state machine** for a Pi-agent live in **S2 §2.4** — this flow stops at dispatch.

Flow 4 — Memory write & promotion (TRD §4.5, blueprint §4.2)

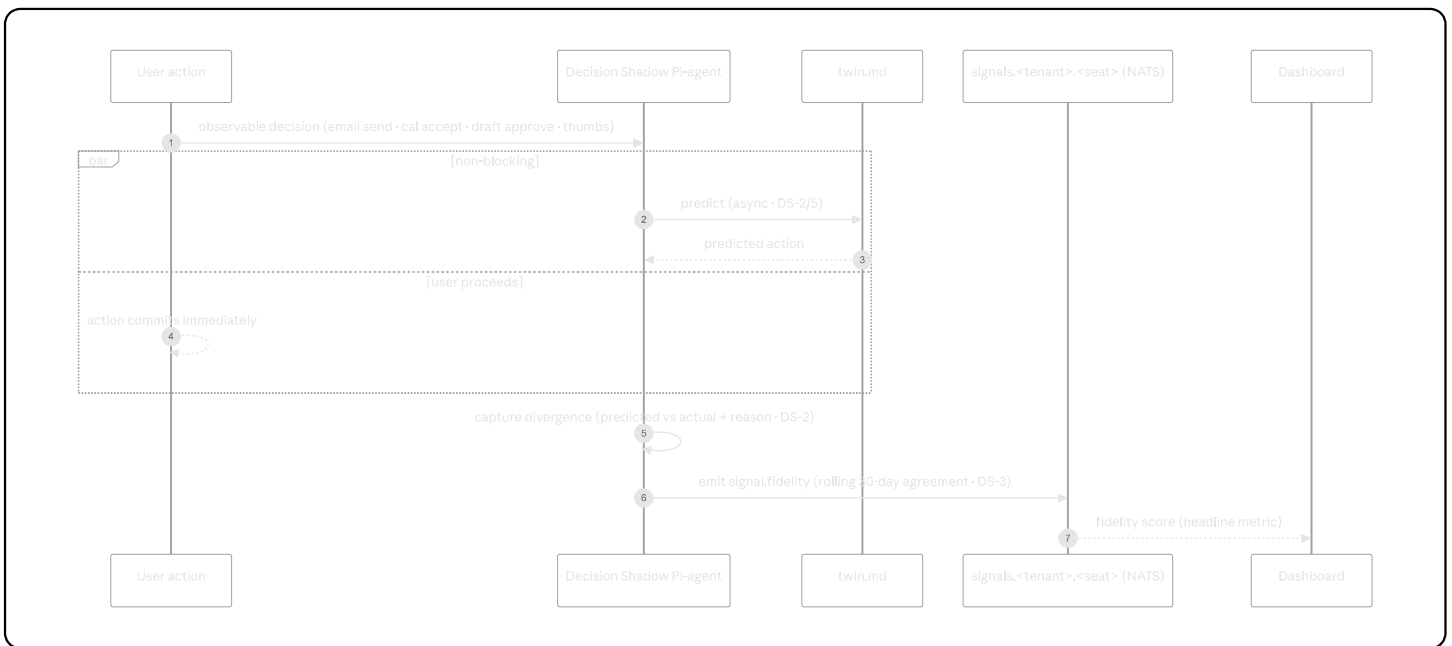
A fact graduates from "I noticed this" to "the company knows this" **only by passing five gates** — the sole path data leaves a seat's private surface.



Gate	Check
1	Confidence threshold — wing-specific minimum bar
2	Redaction — automated PII/secret scan masks values
3	Provenance — source, line, timestamp, origin attached
4	User approval — Promote / Edit / Reject in the Decision Queue
5	30-day rollback — reversible, with a do-not-re-promote marker

Flow 5 — Decision Shadow & fidelity (TRD §4.10)

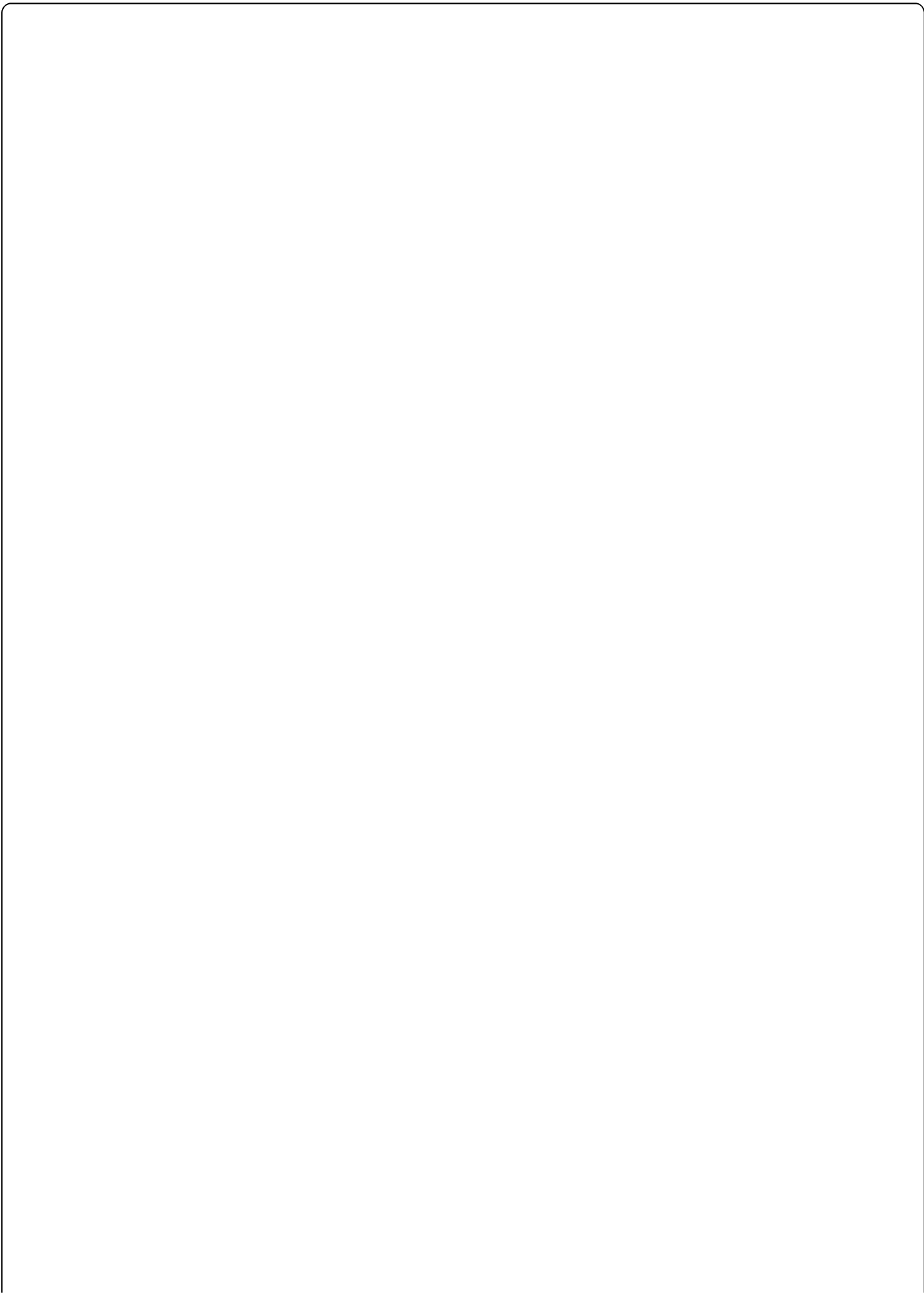
On every observable decision, the **Decision Shadow Pi-agent** runs the twin's prediction **in parallel — never blocking** — and feeds divergence into the fidelity score.

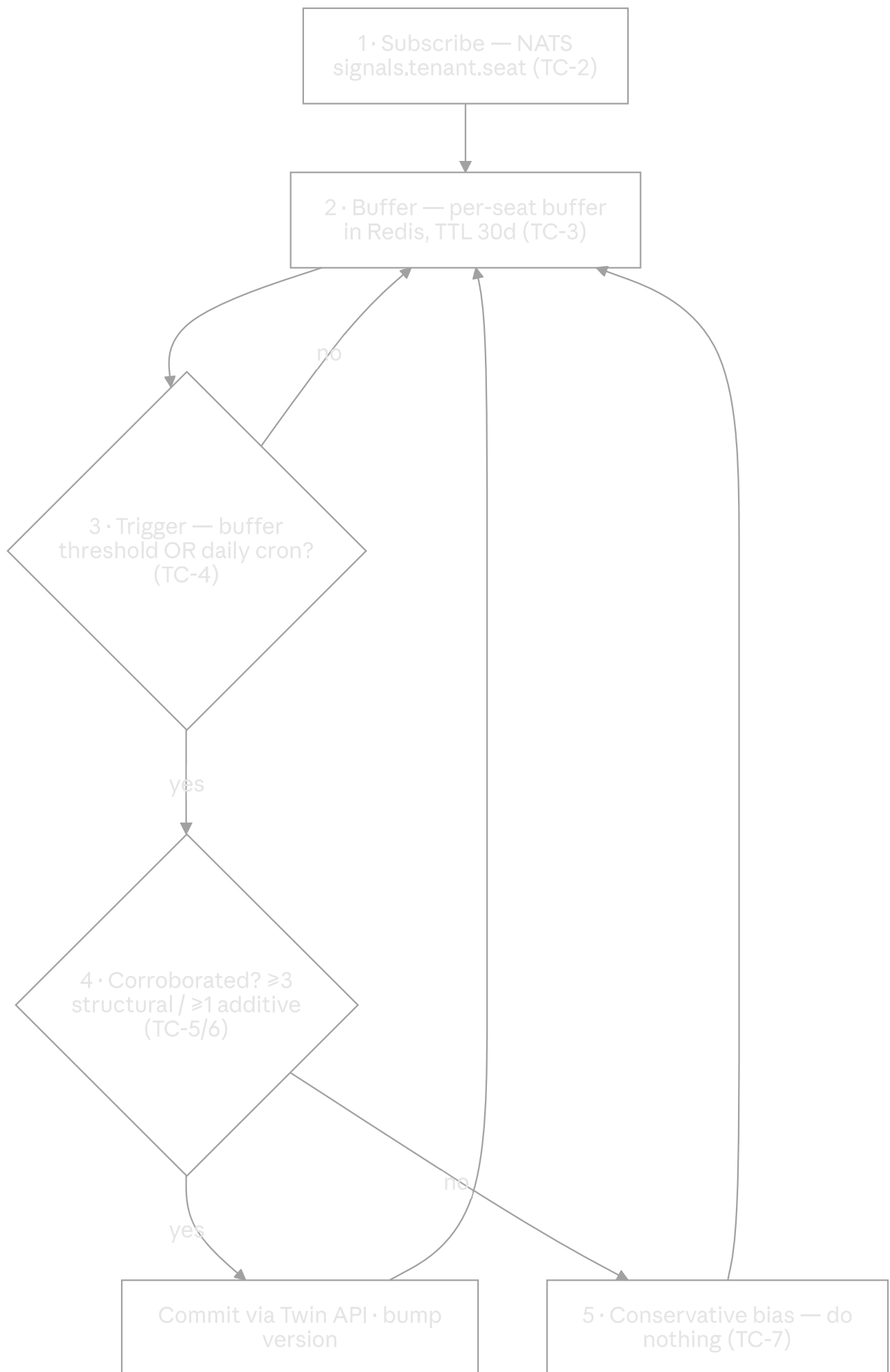


- **DS-4 classes** — V1 tracks **communicative** (text-comparable) + **selective** (binary/n-ary); generative tracked qualitatively only.
- **DS-6 budget** — ≤ 50 shadows/day per seat, prioritized by class — cost-bounded by design.
- **Fidelity** — $\text{agreement_rate_30d} = (\text{class-weighted agreed shadows}) / (\text{scored shadows})$ over the trailing 30 days. Target **0.65 @ D90** → **0.75 @ D180**.

Flow 6 — The twin evolves (TRD §4.9)

The **Twin Curator Pi-agent** turns a signal stream into careful, corroborated, versioned edits — and never overwrites lightly.





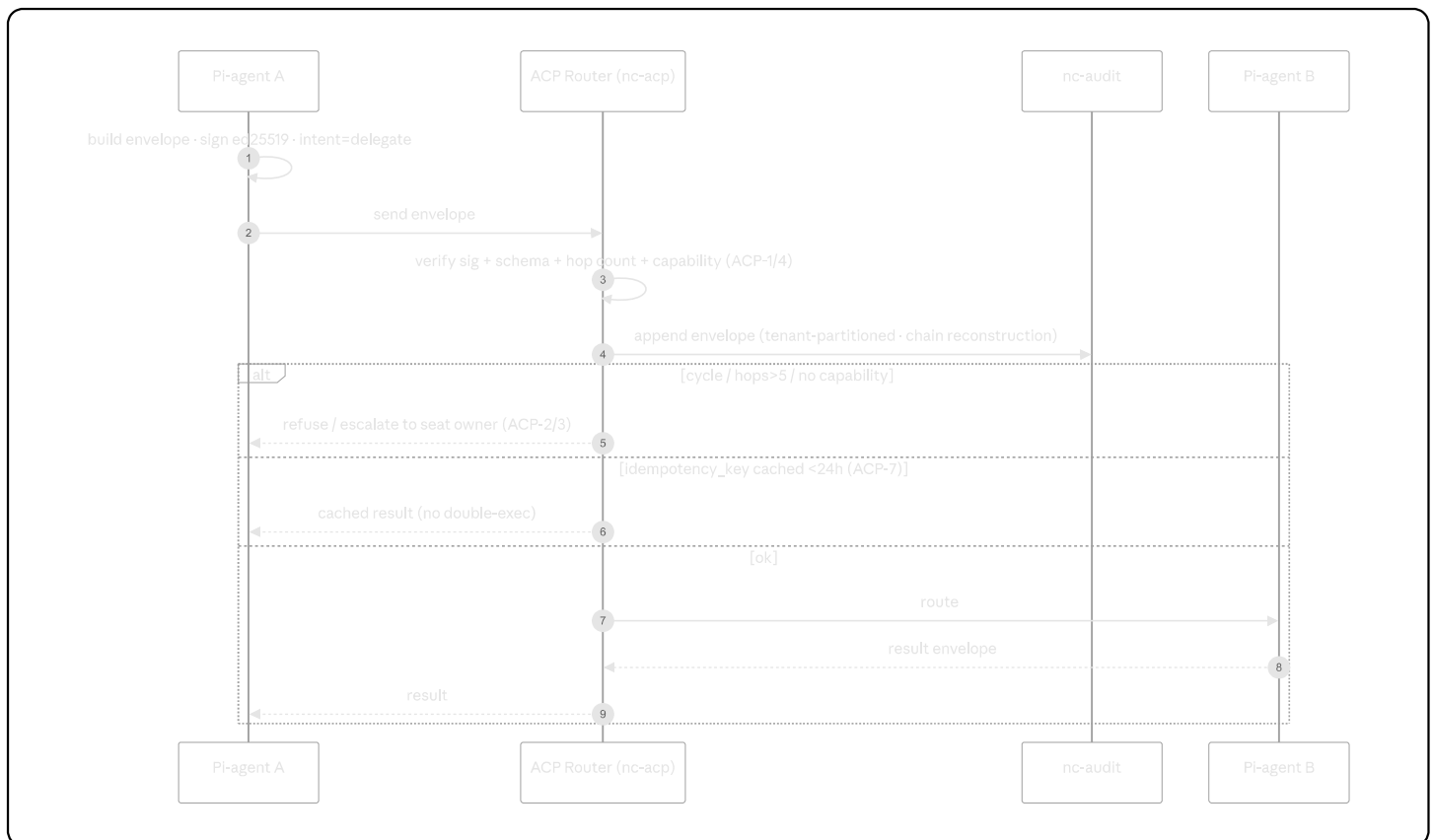
override pattern

Repeated overrides →
re-tune prompt on
dashboard

Drift, not silent rewrite: when the user repeatedly overrides the twin, drift is **surfaced** as a **re-tune prompt** — the system asks, it does not assume.

Flow 7 — Inter-agent delegation (TRD §4.7)

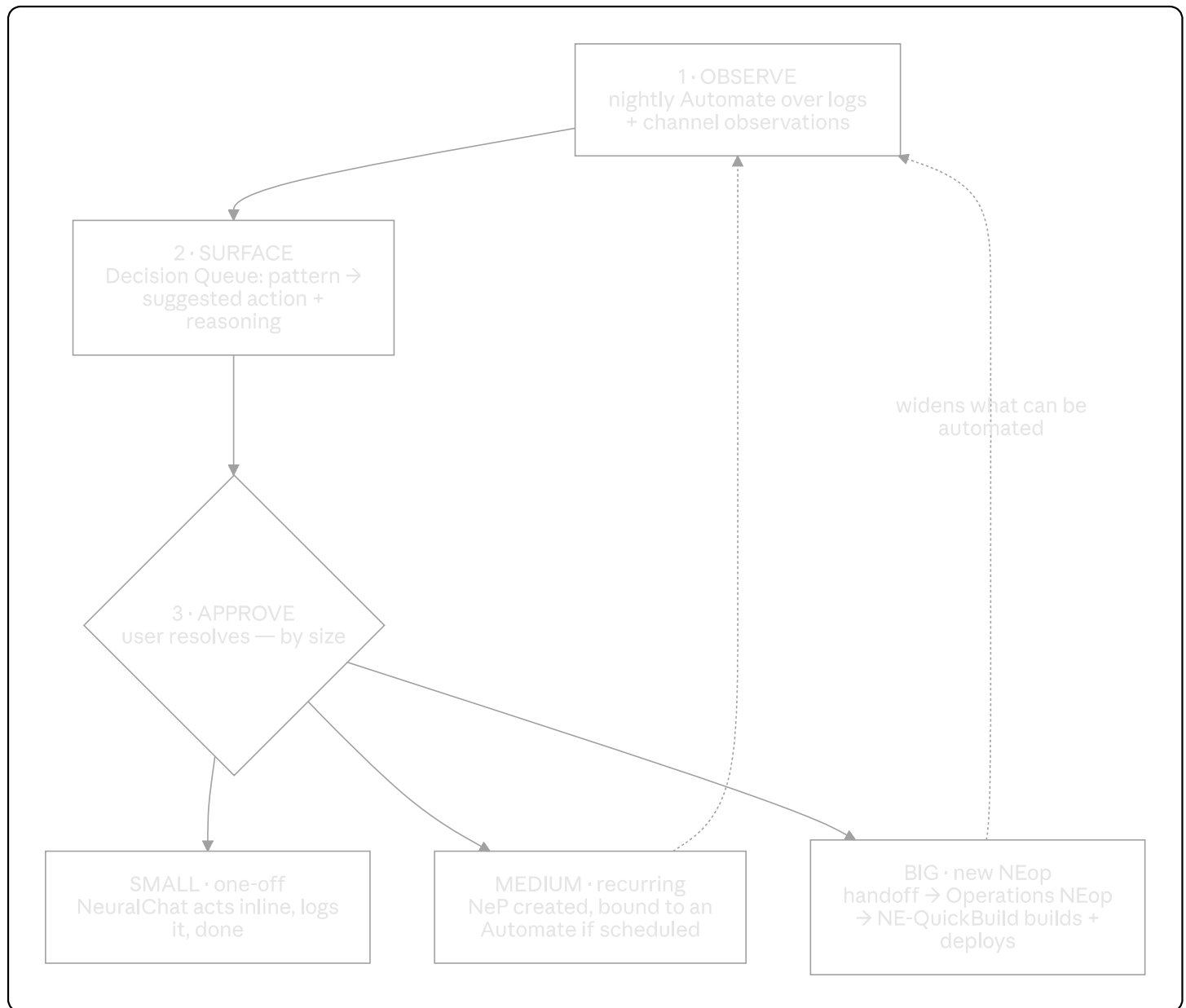
When one Pi-agent needs another, it sends a **signed ACP envelope**. The router is the single chokepoint where every safety check fires before routing (verification pipeline detailed in S2 §2.7).



Cross-tenant delegation is **V2 only**, and only when **both** tenant policies have explicitly opted in (ACP-5); otherwise rejected at the gateway.

Flow 8 — The automation flywheel (blueprint §7)

The product's compounding loop: NeuralChat observes and surfaces; the user approves; small work runs inline, recurring work becomes a NeP, big work becomes a new NEop.

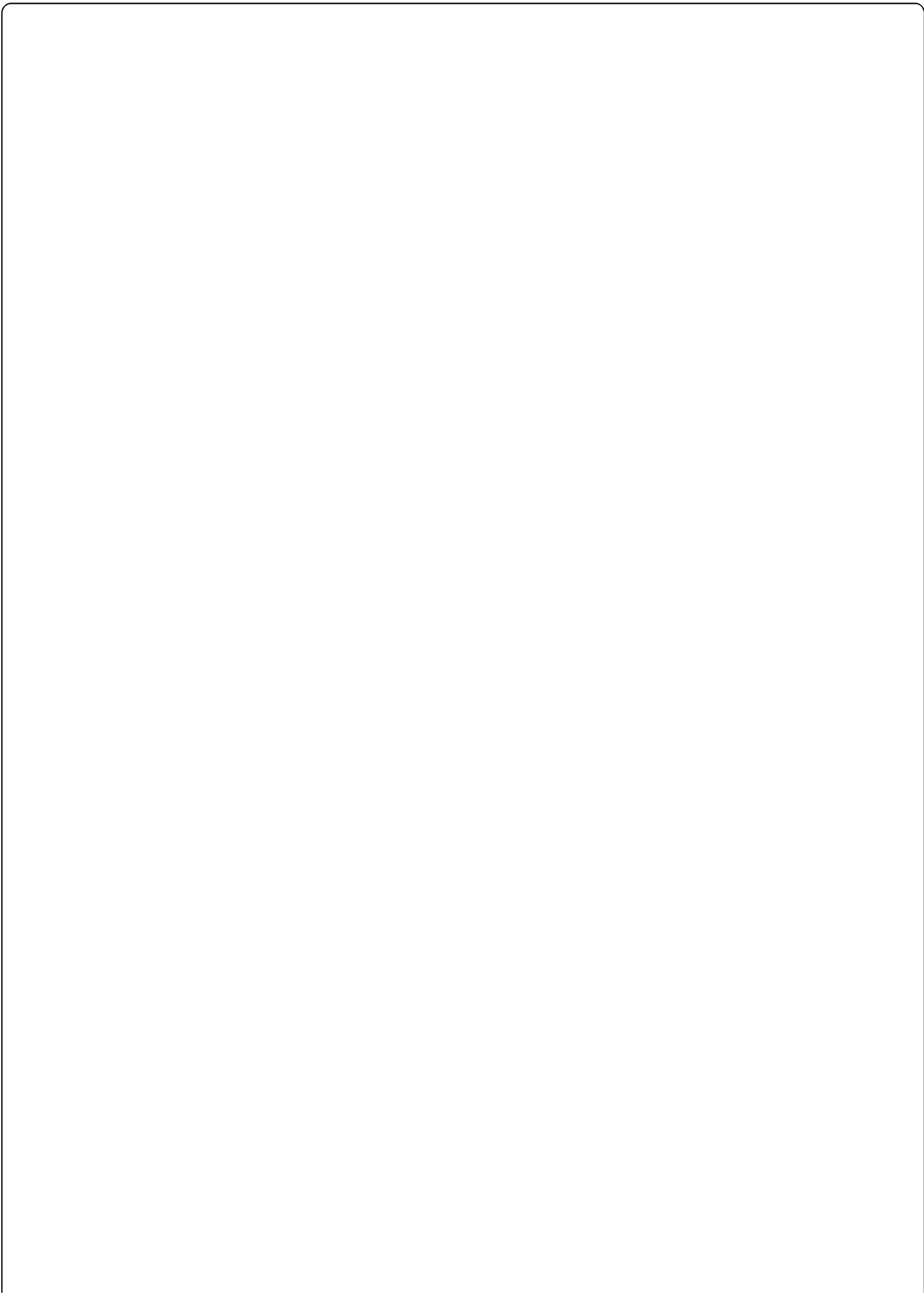


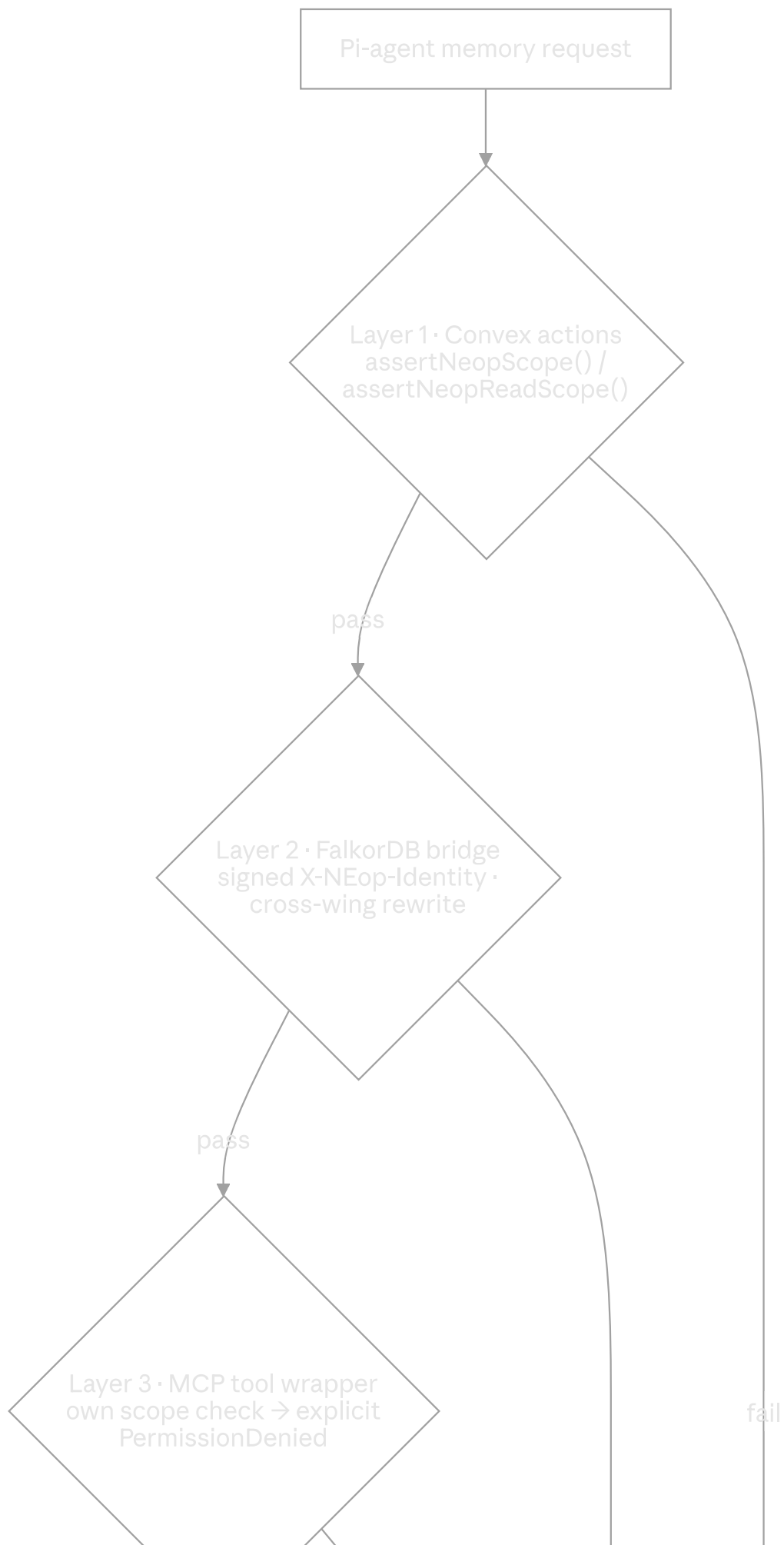
V0 scope: NeuralChat sends the handoff message and stops — the receive side (interview-to-spec, spec-to-NEop, integration receipt) is V1, gated on the **Integration-Receipt Allowlist**.

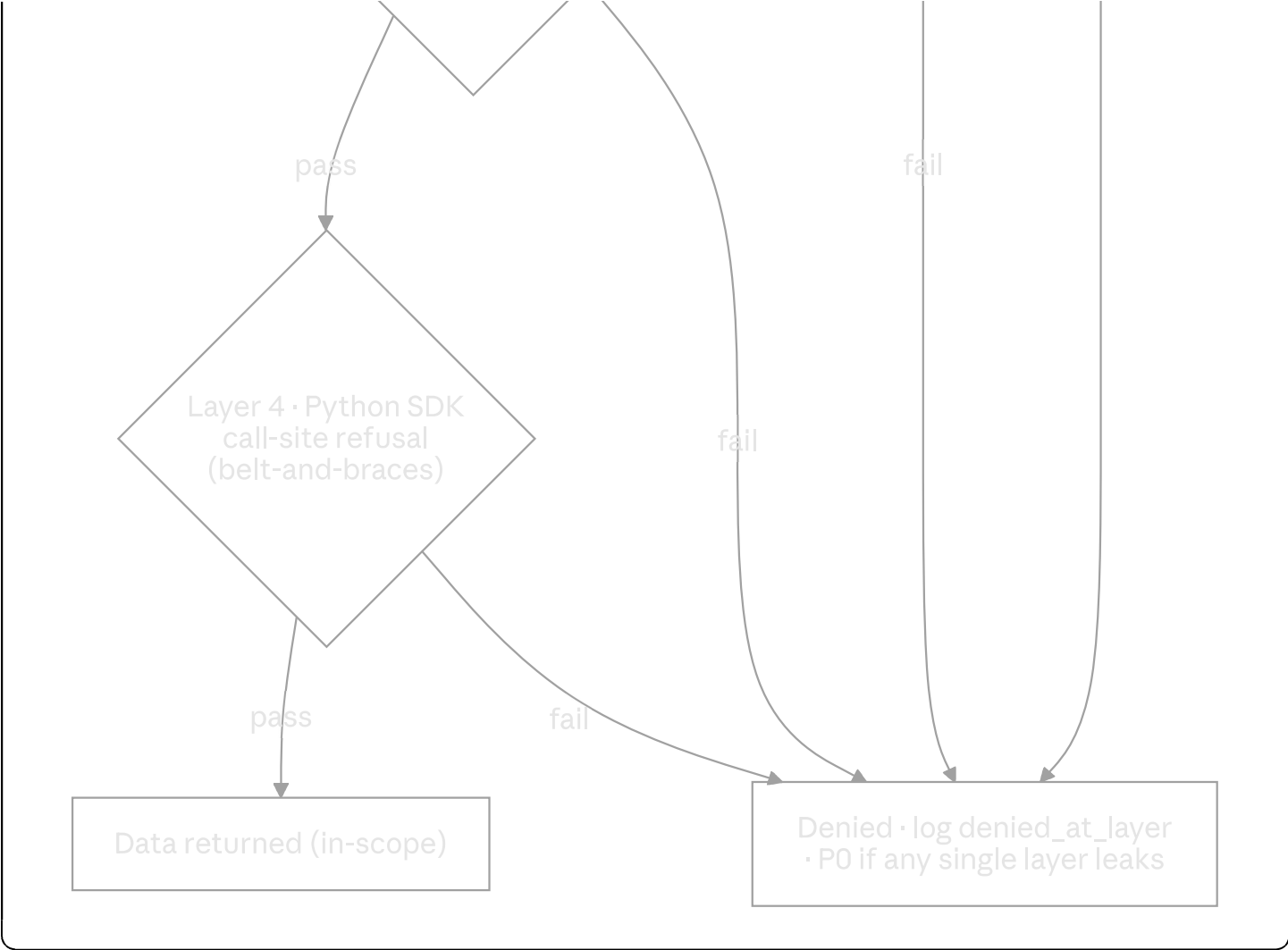
3.2 Security (TRD §7)

3.2.1 Tenant isolation — four-layer ACL (TRD §7.5, blueprint §4.2.4)

"Scoped Pi-agent can read company wings but only write its own room" is enforced **independently at four layers — no layer trusts another alone.**







Test invariant: CI runs an ACL suite against **all four** layers. A bypass at any single layer is a **P0 bug**; every denied operation logs `denied_at_layer` to catch a layer silently passing.

3.2.2 Auth · authz · encryption · secrets

Domain	Spec
Authentication	Matrix-token (V1) → SSO/OIDC (V2); service-to-service mTLS in VPC + per-service keys ; tool creds in AWS Secrets Manager — never env vars ; ACP envelopes Ed25519-signed , keys rotate quarterly
Authorization	Three role tiers — owner · admin · member ; OPA/Rego on every gated action; PALACE client enforces <code>(seat, tenant)</code> on every op
Encryption	TLS 1.3 all traffic; at rest AWS KMS per-tenant CMK (V2) ; Matrix Megolm E2EE by default; twin file seat-derived key (V2)
Audit (SEC-12... 15)	Every state change logs who · when · what · on whom · with what permission · result ; append-only, tenant-partitioned; customer export ≤24h; staff access reason-gated + audited

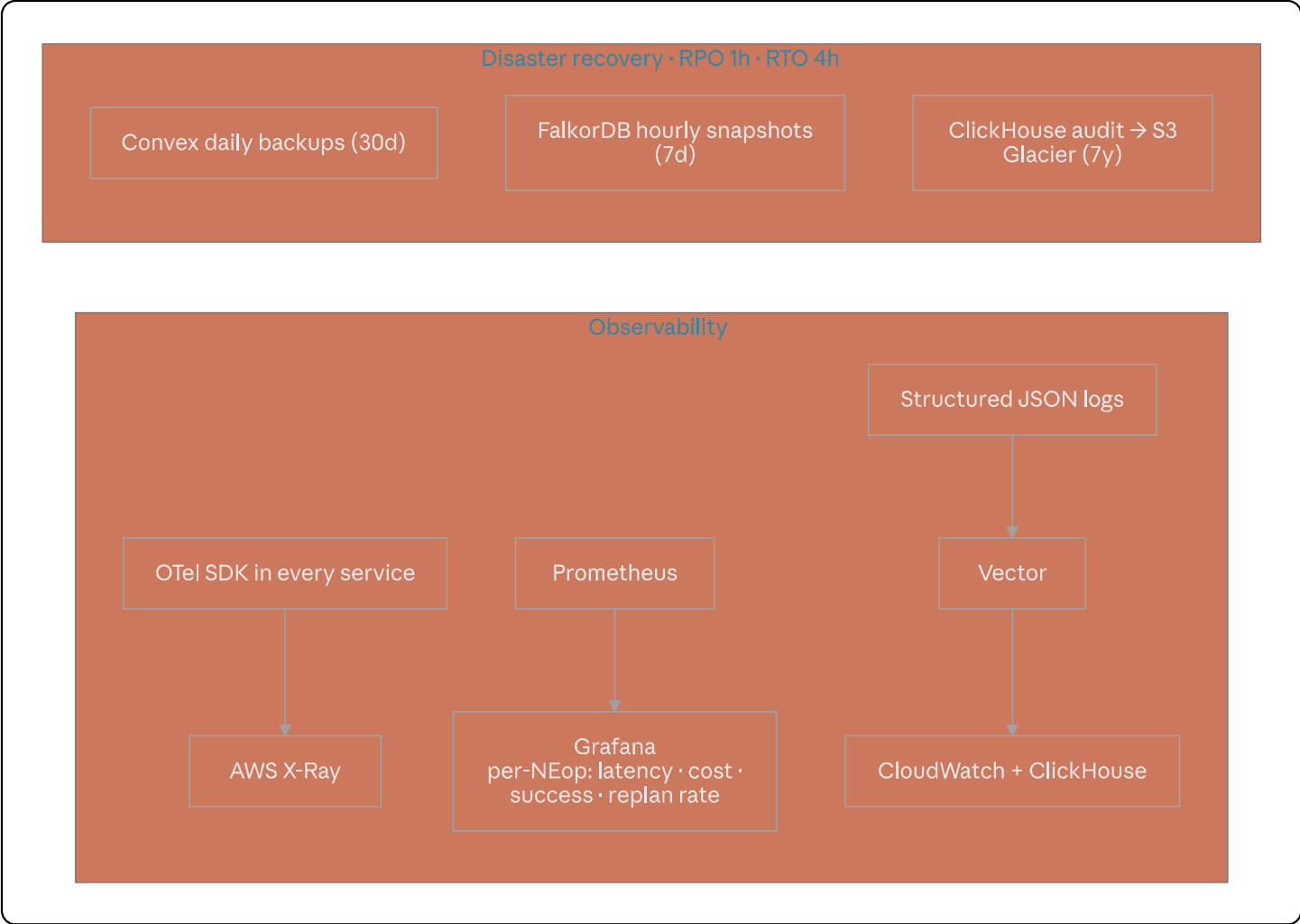
Domain	Spec
Secrets posture	V0 accepts <code>chmod-600 .env</code> (single user). V1 blocker: all secrets → OS keychain via Python <code>keyring</code> before user #2 onboards

3.3 Reliability & operations (TRD §8)

3.3.1 SLOs

SLO	Target	Error budget
Gateway availability	99.9%	43.2 min/mo
Chat round-trip p95	≤ 2 s	5% of requests
Pi-agent run p95	≤ 60 s	5% of runs
PALACE retrieval p95	≤ 1.2 s	5% of queries
Twin write durability	100%	0 — no acked loss

3.3.2 Observability & DR



Incident response: 24/7 PagerDuty for P0/P1 · public status page · post-incident review within 5 business days, drafted by the **Postmortem Drafter** Pi-agent. RPO 1h / RTO 4h for a full tenant restore — **drilled quarterly**.

3.4 Compliance & privacy (TRD §9)

Req	Baseline
COM-1	DPDP Act 2023 (India) — V1 must comply; first tenant is India-based
COM-2	SOC 2 Type 1 — target Q4 2026 (V2)
COM-3	GDPR-ready architecture — must not block EU expansion
COM-4	ISO 27001 — target 2027
COM-5	Data residency — all Indian tenant data stays in <code>ap-south-1</code>

Privacy commitments: decision cloning is explicitly disclosed and consented at onboarding · per-class shadow opt-out (communicative / selective / generative independently) · a user can view and correct their own `twin.md` anytime · on offboarding, personal patterns export per company policy · staff access requires a reason, is audited, and is customer-notifiable.

The ethics line: decision shadowing is the product's most sensitive surface; the mitigation is **transparency as a first-class feature** — explicit opt-in, per-class controls, and a dashboard that always shows what is being tracked.

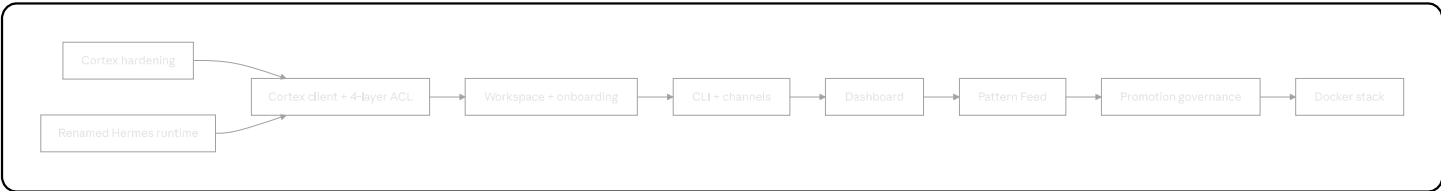
3.5 The build (TRD §10, §15)

3.5.1 Locked stack

Hermes runtime host · **Pi-agents** as execution unit · Convex (SoT) · FalkorDB + Graphiti · Matrix Synapse · TanStack Start · Node 22 · Python 3.12 · ClickHouse (audit) · NATS (events) · Redis (cache) · OPA/Rego · AWS `ap-south-1` · Gemini → Qwen3 embeddings.

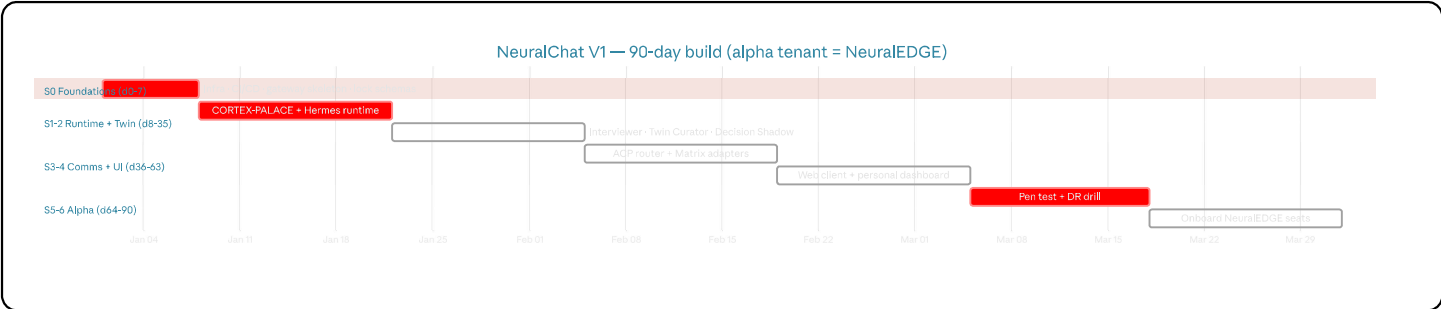
Models: Claude Sonnet for planning · GPT-5.4-mini general · Haiku-class classifier. V2 adds ZAI/GLM-5 as a local fallback.

3.5.2 Critical path — dependency order



The substrate must be correct before any user-visible surface is built. Cortex hardening and the renamed Hermes runtime run in parallel; everything downstream waits on the Cortex client + four-layer ACL.

3.5.3 90-day schedule



Sprint	Days	Deliverable
S0	0-7	Foundations — infra, CI/CD, gateway skeleton, lock schemas
S1-2	8-35	Runtime + Twin — PALACE, Hermes runtime, Interviewer, Curator, Shadow
S3-4	36-63	Comms + UI — ACP, Matrix, web client, dashboard
S5-6	64-90	Alpha — pen test, DR drill, onboard NeuralEDGE seats

3.5.4 V1 acceptance — Day 90

- NeuralEDGE tenant onboarded — ≥ 5 seats, `twin.md` version ≥ 5
- ≥ 3 NEops in regular use per seat
- Average twin fidelity ≥ 0.65 across active seats
- Chat round-trip p95 ≤ 2 s · Pi-agent run p95 ≤ 60 s in production
- **Zero tenant-isolation violations in 30 d** — verified by audit replay
- All 6 meta-NEops live · personal dashboard live · pen test passed

3.5.5 Top risks & open questions

- **Fidelity may not hit 0.75 in 90 days** — mitigated by conservative ramp, Shadow tuning, the ≥ 3 -signal rule.
- **Codex review hardened 32 findings** — RCE-equivalent auto-apply (now an allowlist), unspecified ACL (now 4-layer defense in depth), coarse permissions (now 90-day T3 leases).
- **8 open questions lock between Day 14-30** — Synapse topology · event bus · fidelity algorithm · twin commit model · (+4) — each locked before it blocks its dependent sprint.

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